

Embedded System Architecture - CSEN 701

Module 3: Embedded Hardware **Lecture 07: Actuators & PWM**

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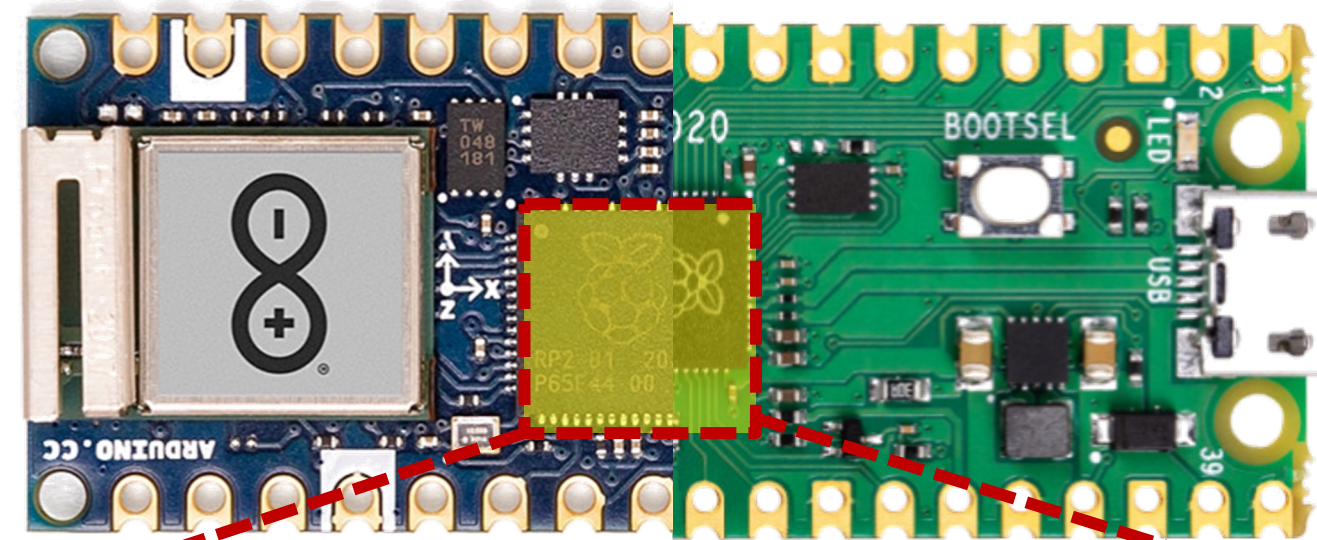
- Output Devices
- DC Motors
- PWM
- Servo Motors

Where are we?

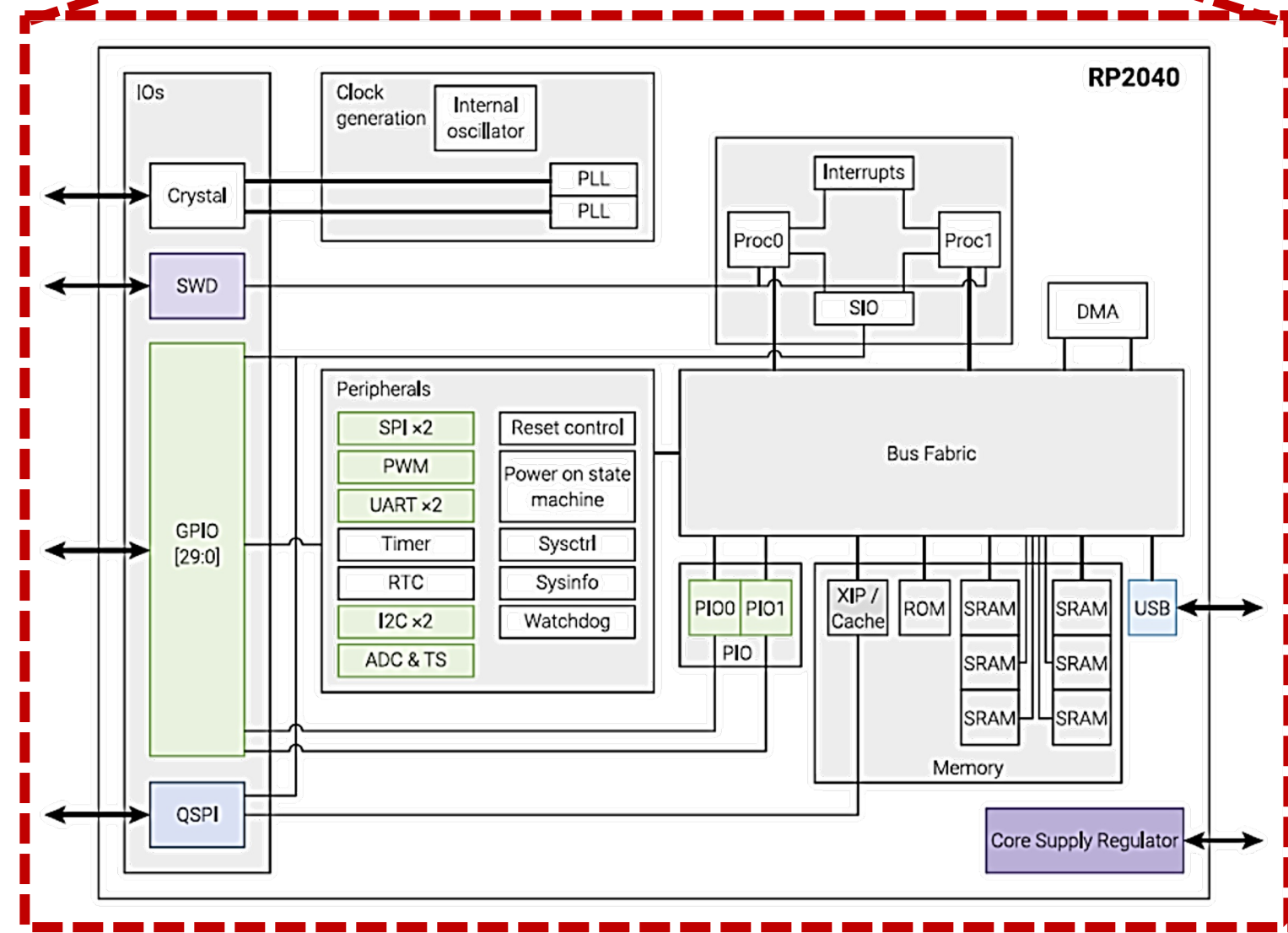
An Embedded System

Physical
System Input
“Sensor”

Microcontroller



Physical
System Output
“Actuator”



- What is the **nature** of the **I/O Signals**? ✓
- How to make sure that the microcontroller understands these signals (**Interfacing techniques**)? ✓
- What are the **sensors** and **actuators**? *Let's go* ✓
- How to **read** the sensor? ✓
- How to **control** the actuators? *Let's go*

Definition

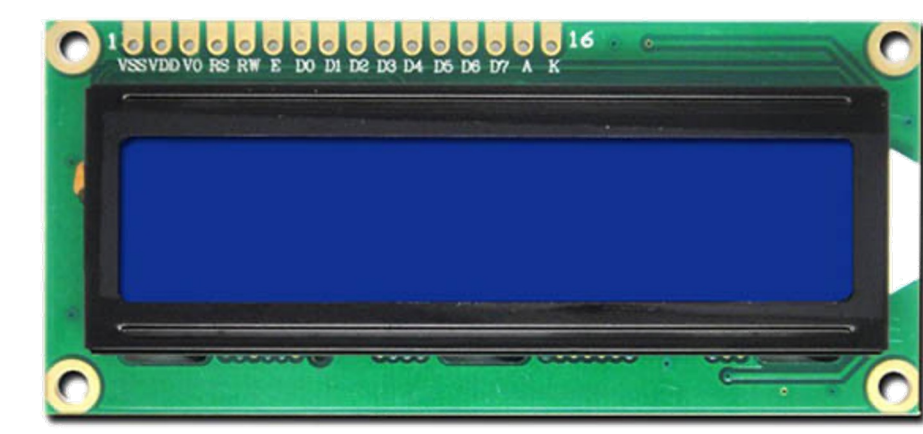
- Output devices in embedded systems are **components** or **peripherals**.
- These devices are responsible for providing information or responding to commands by producing a **physical output**.
- These devices play a crucial role in **communicating with the external environment** or the user.

Examples



LEDs
(Light Emitting Diodes)

LCD Displays
(Liquid Crystal Displays)



7-Segment Displays

OLED Displays
(Organic Light Emitting Diodes)

E-ink Displays

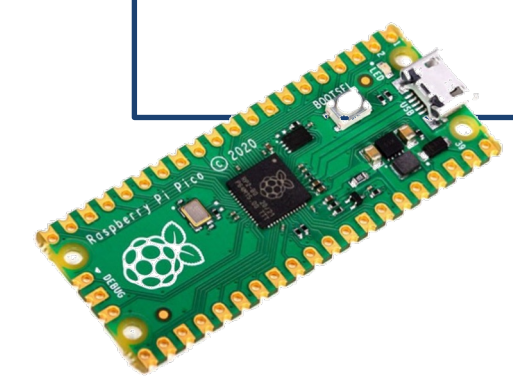
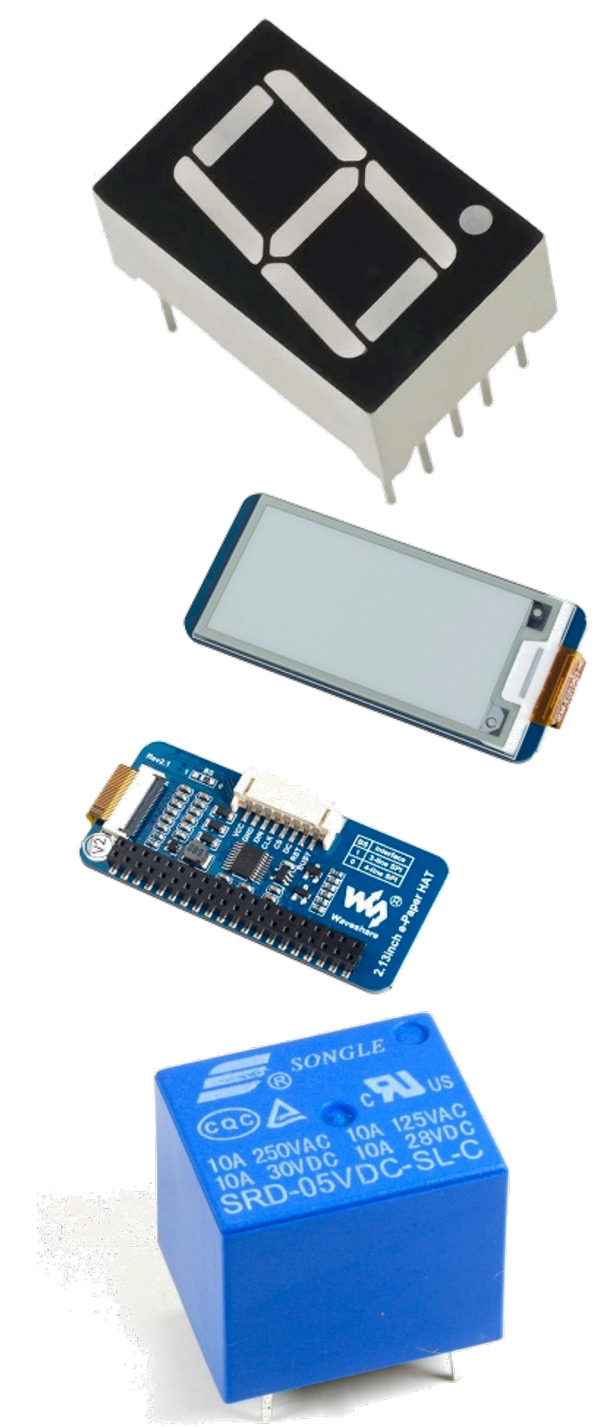
Buzzer or Speaker

Relays

Actuators

Digital Communication
Interfaces

Wireless Communication
Modules



DC Motor



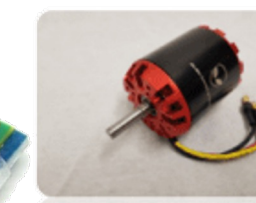
DC Gear Motor



RC Servo motor



Stepper motor



BLDC Motor



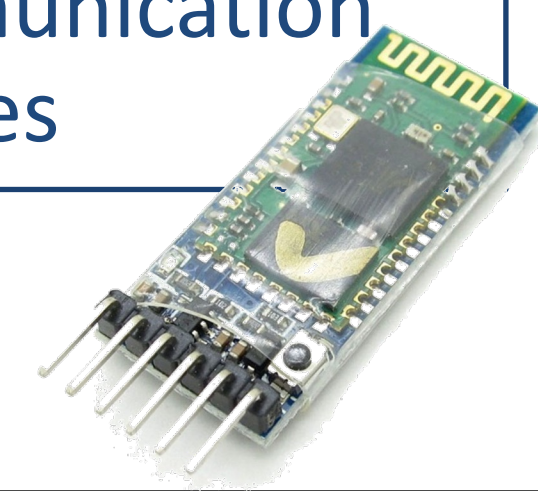
Smart Servo motors



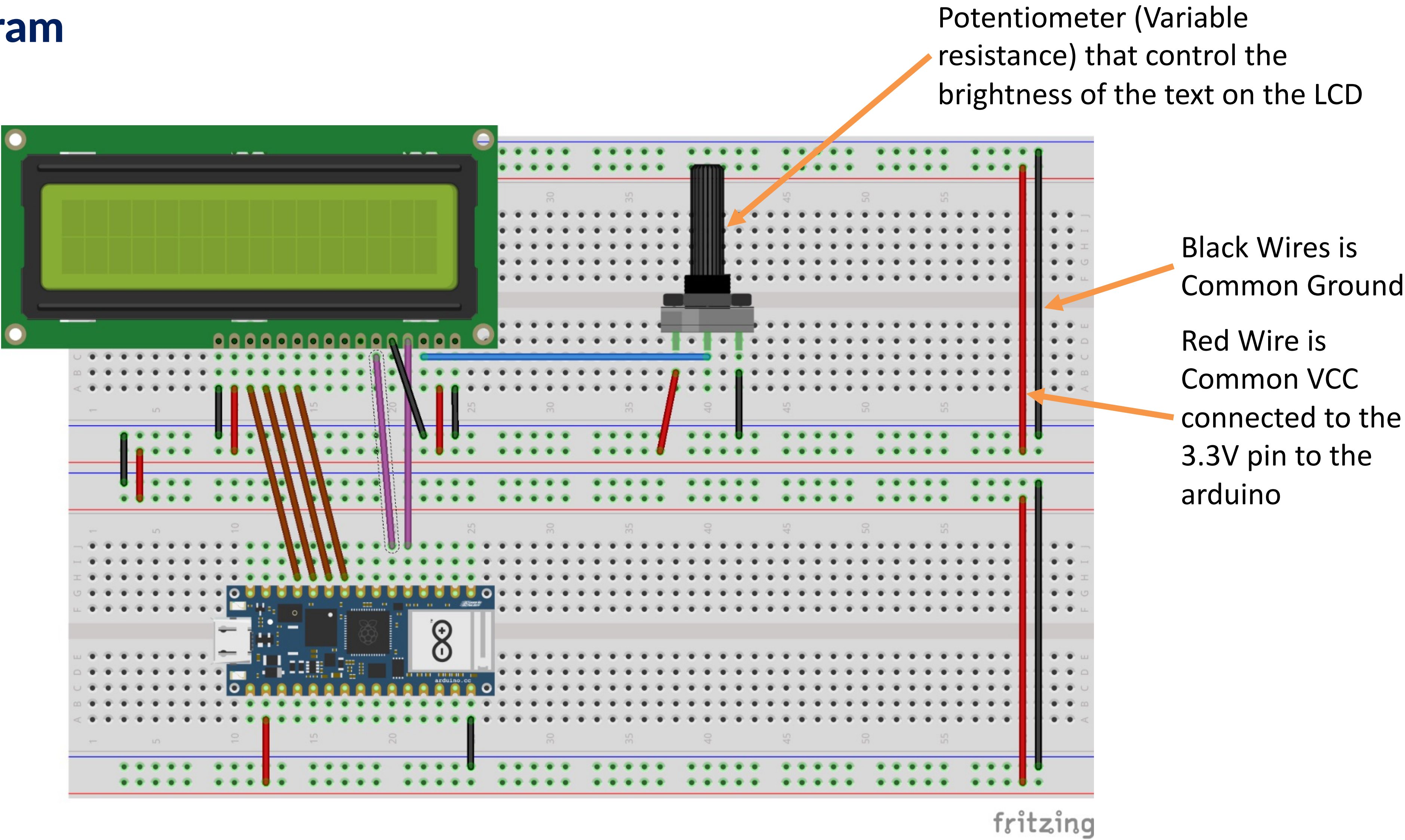
Harmonic drives



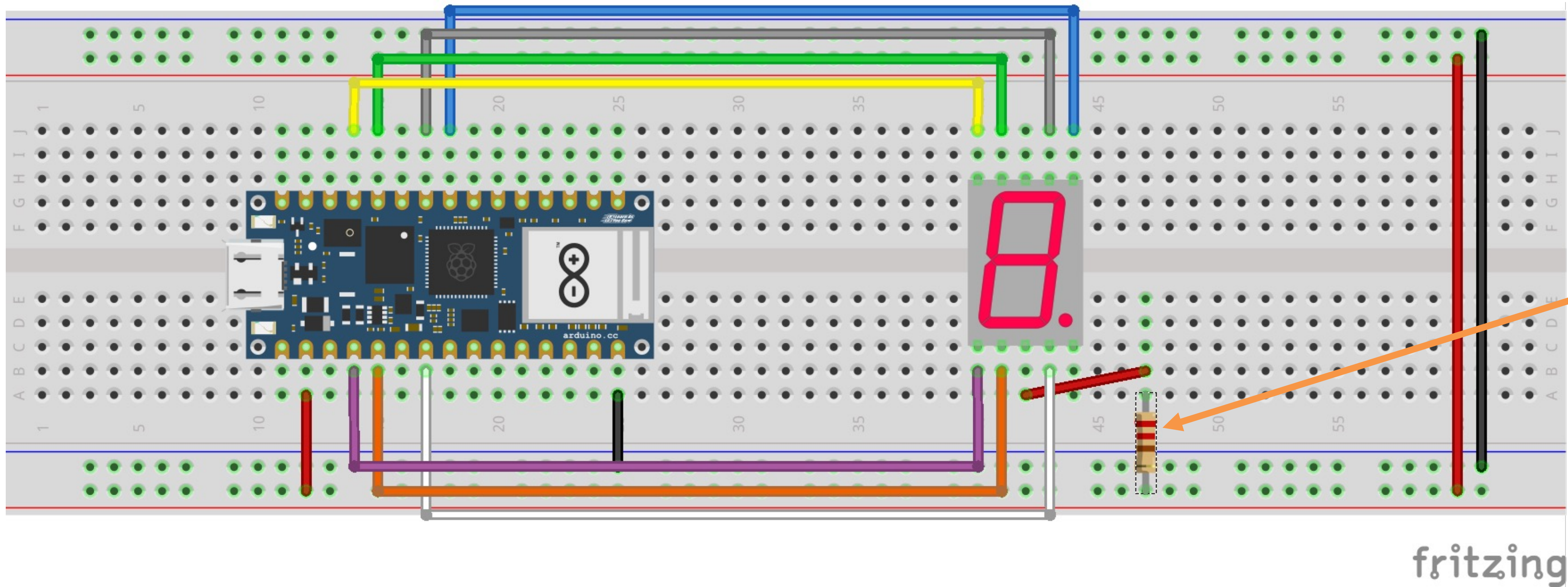
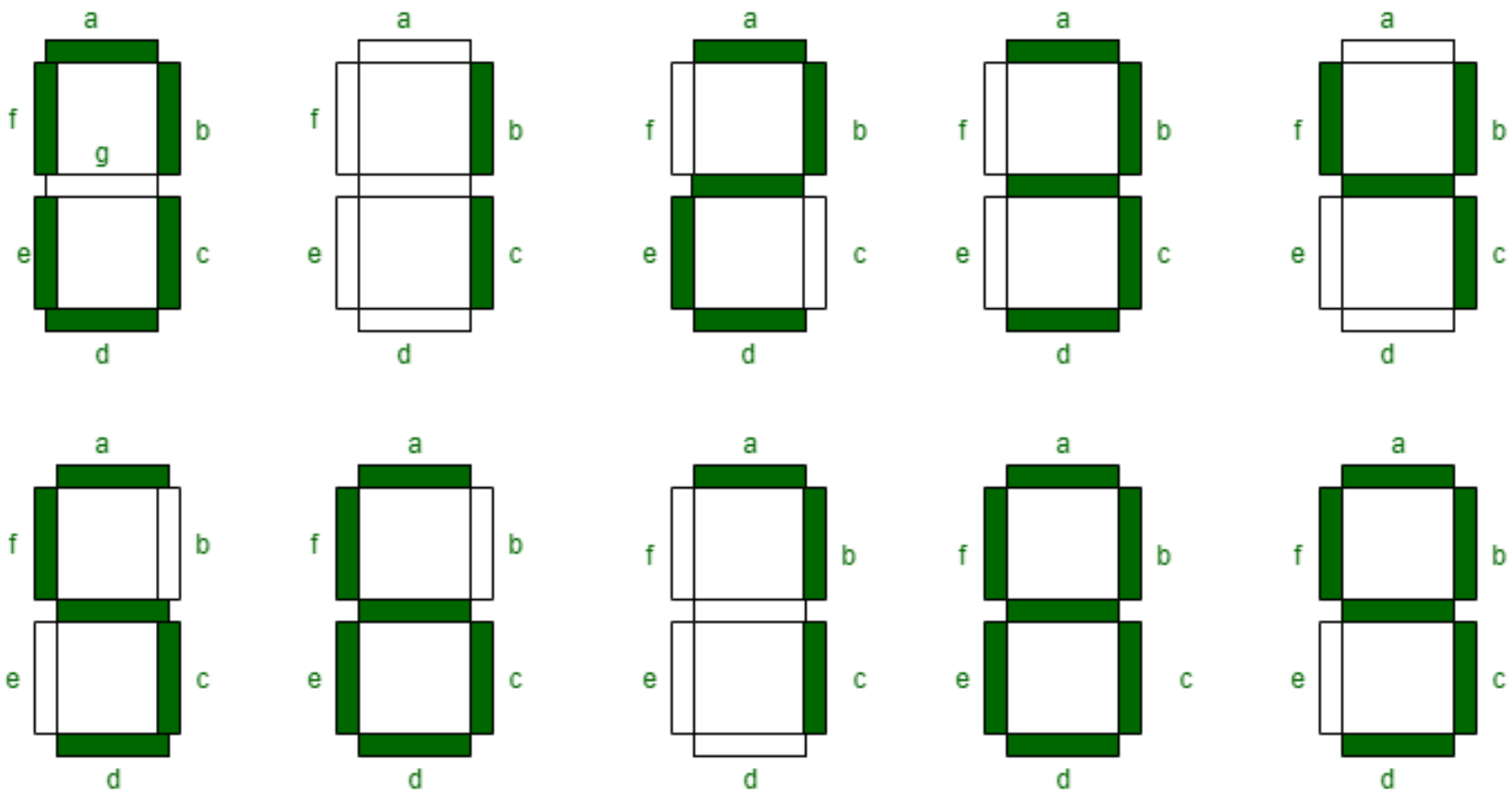
Linear electric actuator



LCD Wiring Diagram



7-Segment Wiring Diagram



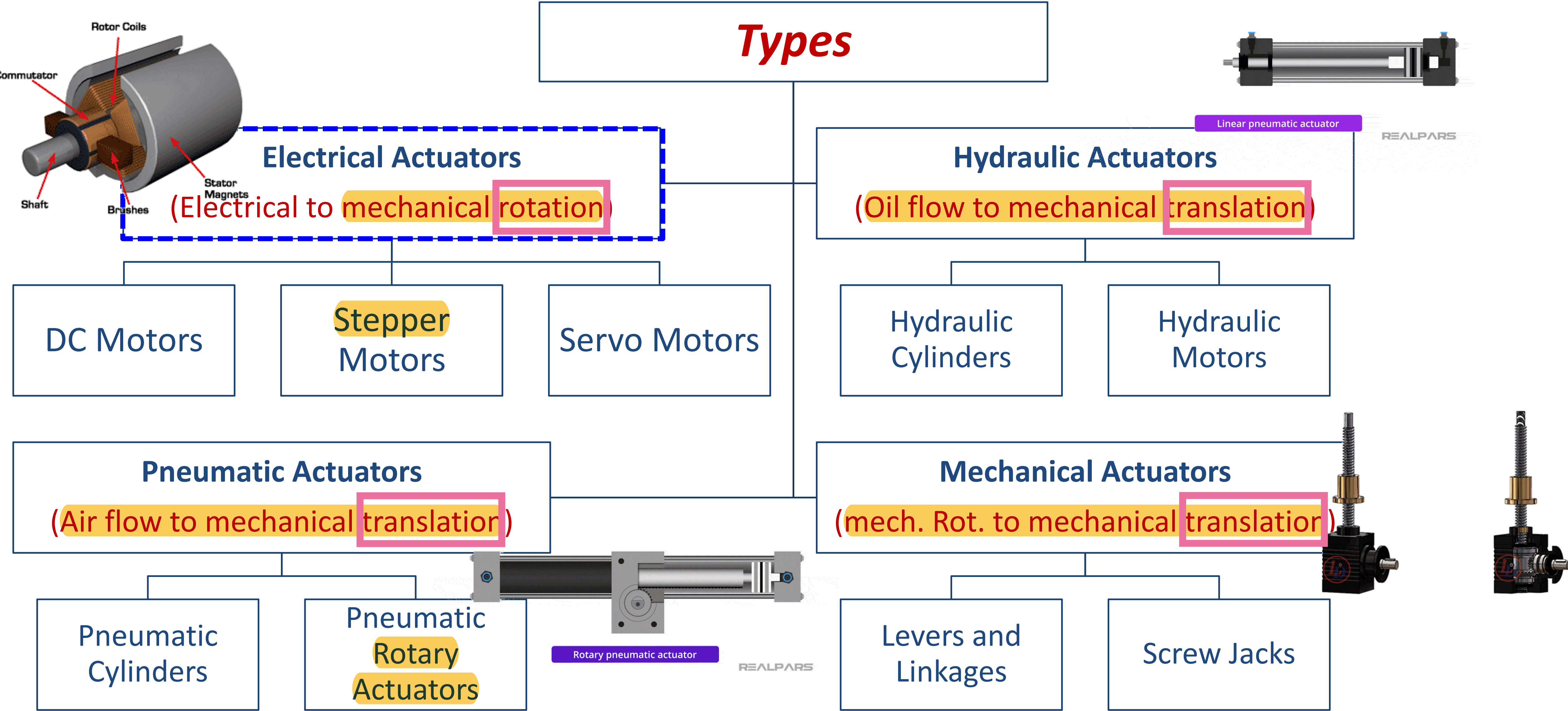
Pull up resistance since it is connected to the Vcc, which means that in order to turn on one of the 7-segments, you need to pull down the resistance which means setting the value to low

Definition

- Actuators are devices or components that are responsible **for converting an input signal or energy into a physical action or motion.**
- They are crucial in various fields, including engineering, automation, and robotics, where they play a pivotal role in **moving, controlling, or manipulating mechanical systems.**

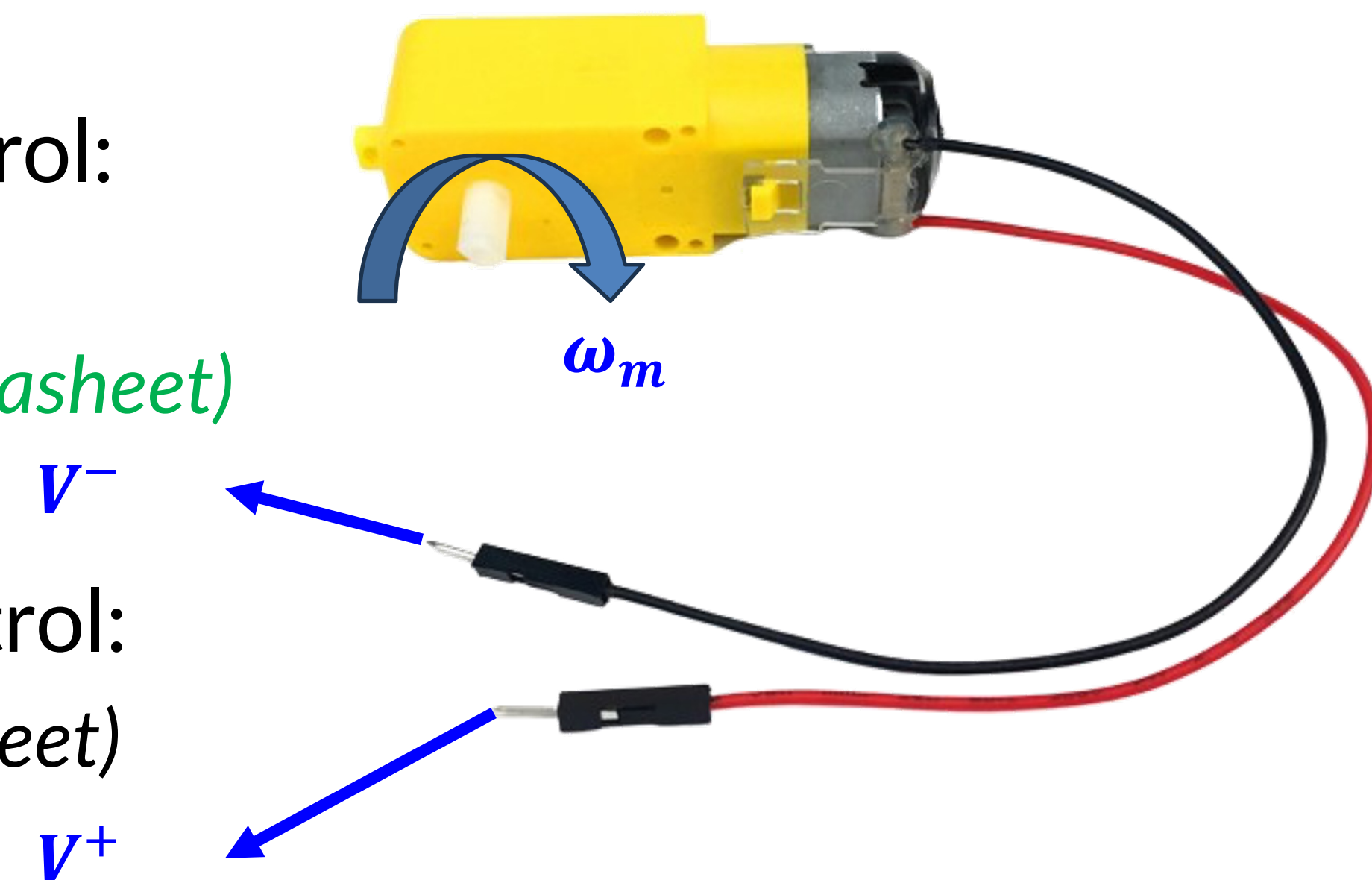
Functionality and Characteristics

- **Movement:** The primary function of actuators is to produce controlled motion. This motion can be linear or rotational.
- **Control:** Actuators are used to control various parameters, such as position, speed, force, or torque.
- **Automation:** They enable the automation of tasks that would be challenging or impractical for humans to perform repeatedly and accurately.
- **Feedback Control:** In many applications, actuators are paired with sensors to create closed-loop control systems. This allows for real-time monitoring and adjustment of the actuator's behavior based on feedback from the system.
- **Energy Conversion:** Actuators often convert one form of energy into another. For example, electrical actuators convert electrical energy into mechanical motion.



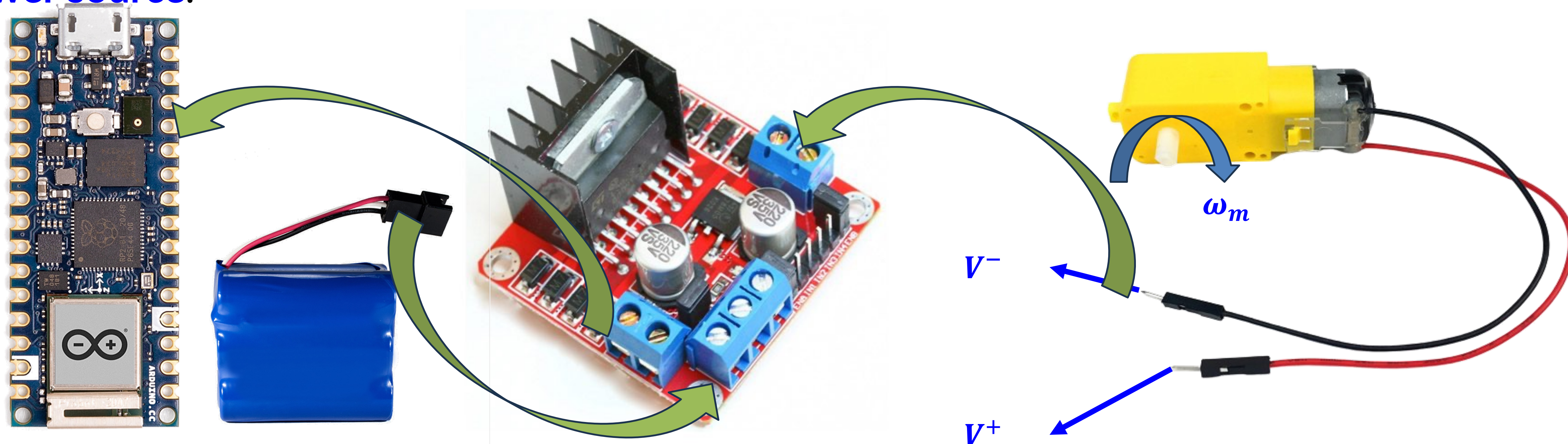
DC Motor: Connection with μC

- It is an analog electric actuator that converts electrical energy into **mechanical rotary motion**.
- It has **2 wires V^- and V^+** .
- By **changing the voltage** given to the motor, you can control:
 1. **Direction of rotation (CW or CCW)**
 2. **Speed of rotation** $\omega_m = K_v \times V_m$ (K_v is voltage constant $\rightarrow$ datasheet)
- By **changing the Current** given to the motor, you can control:
 1. **Torque produced** $\tau_m = K_\tau \times I$ (K_τ is torque constant $\rightarrow$ datasheet)

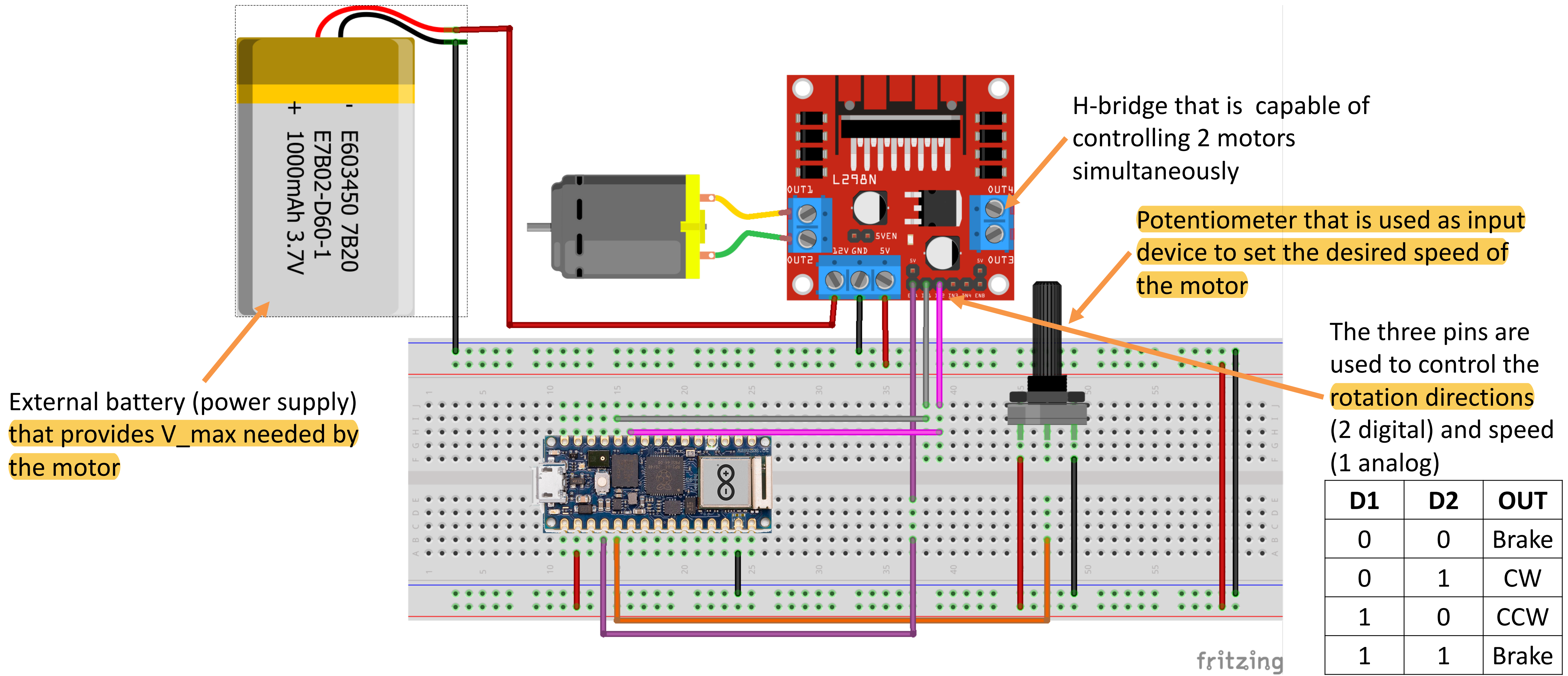


DC Motor: Connection with μC

- Usually, a DC motor works in **high maximum voltage** and drain **high current**.
- Therefore, we do **NOT** connect our DC motor directly to the microcontroller for its protection.
- Instead, an extra intermediate component is used known as an **H-Bridge** with **an external power source**.



DC Motor: Connection with μC

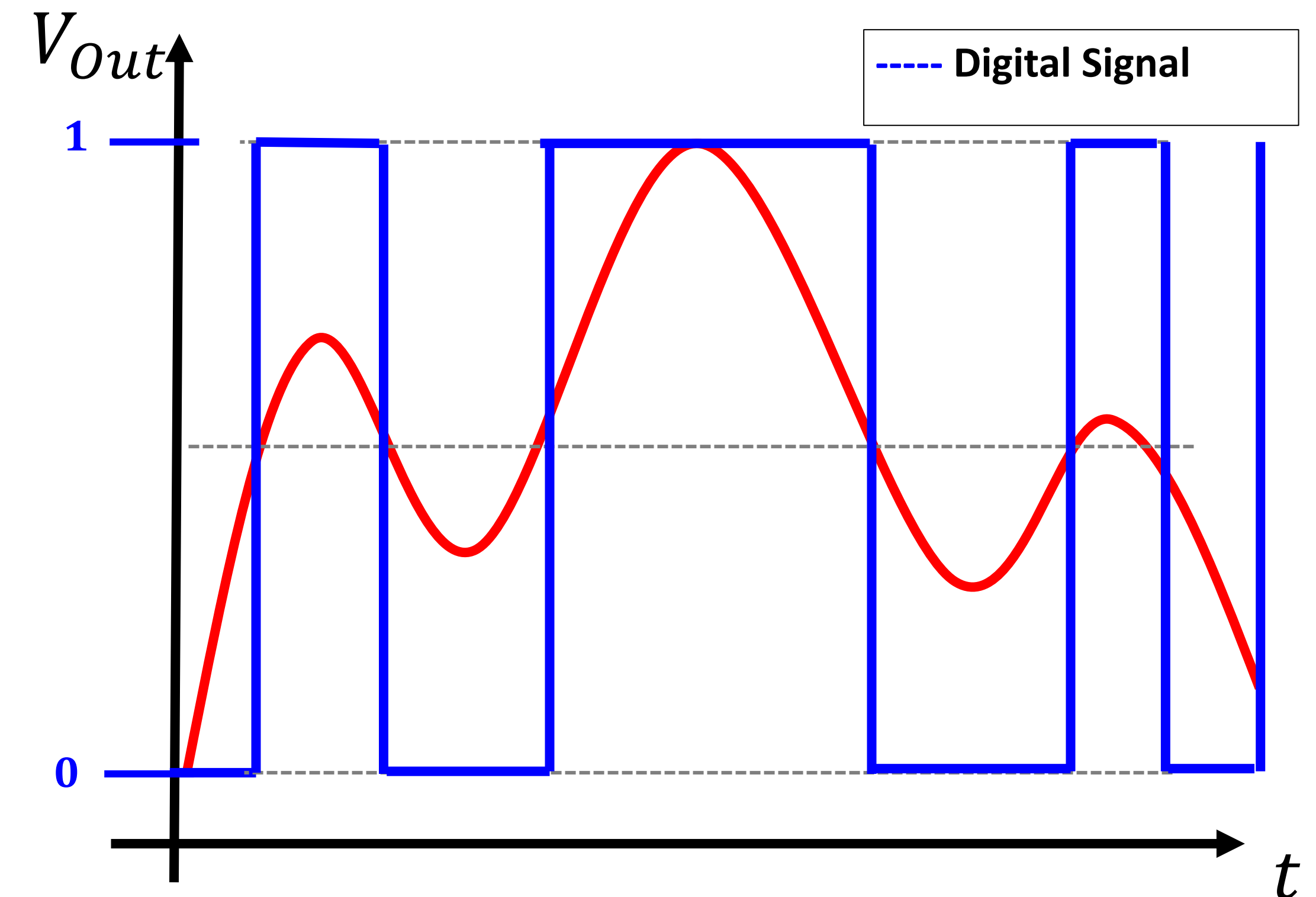
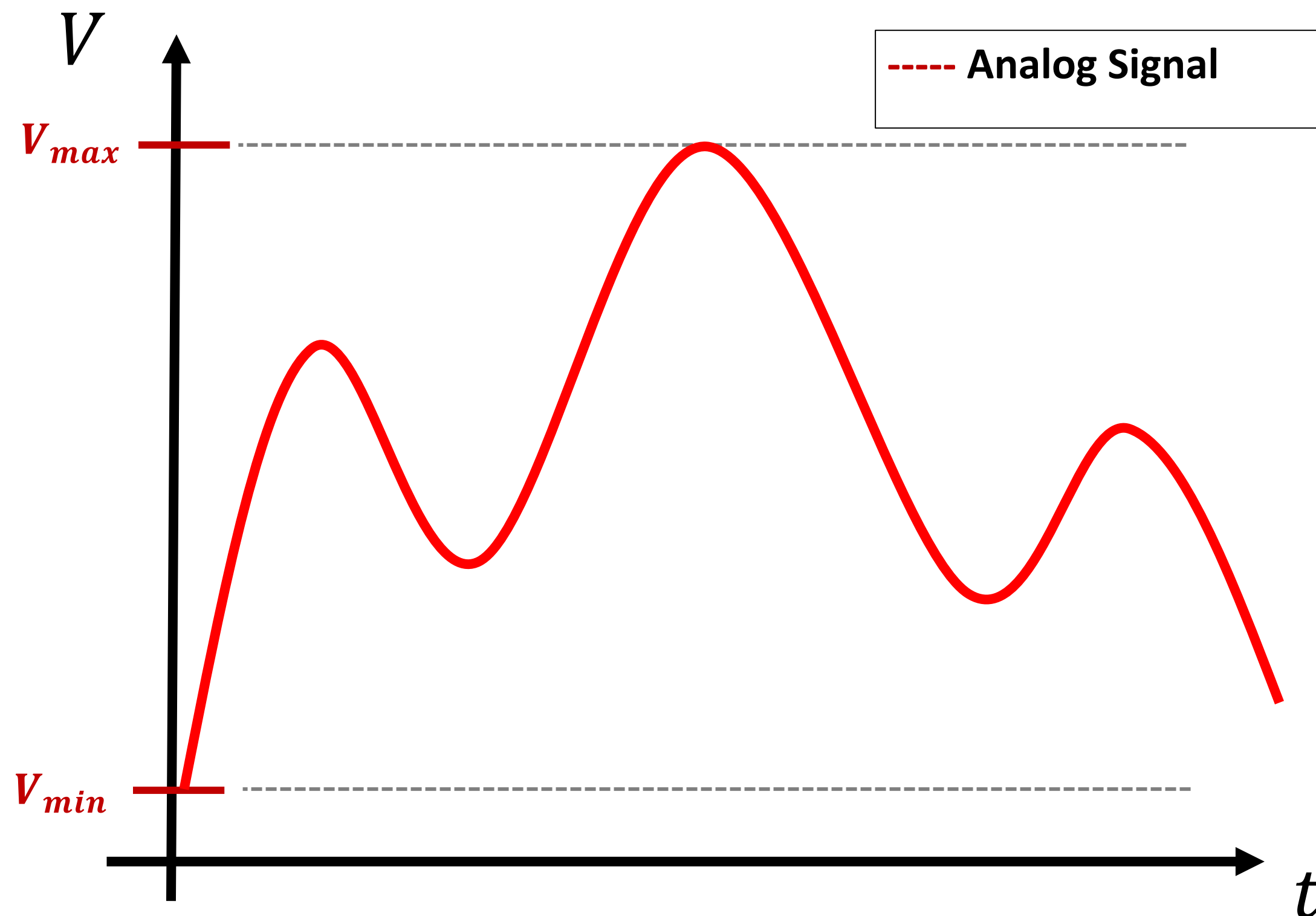


DC Motor: Speed Control

- As previously mentioned, in order to change the speed of the motor ω_m , we should change the motor voltage V_m .
- But before we do that, let's again remember the signals concept:

Note that: 0 means V_{min} and 1 means V_{max} as per the quantization concept.

Note that: the signal is also sampled in the time domain

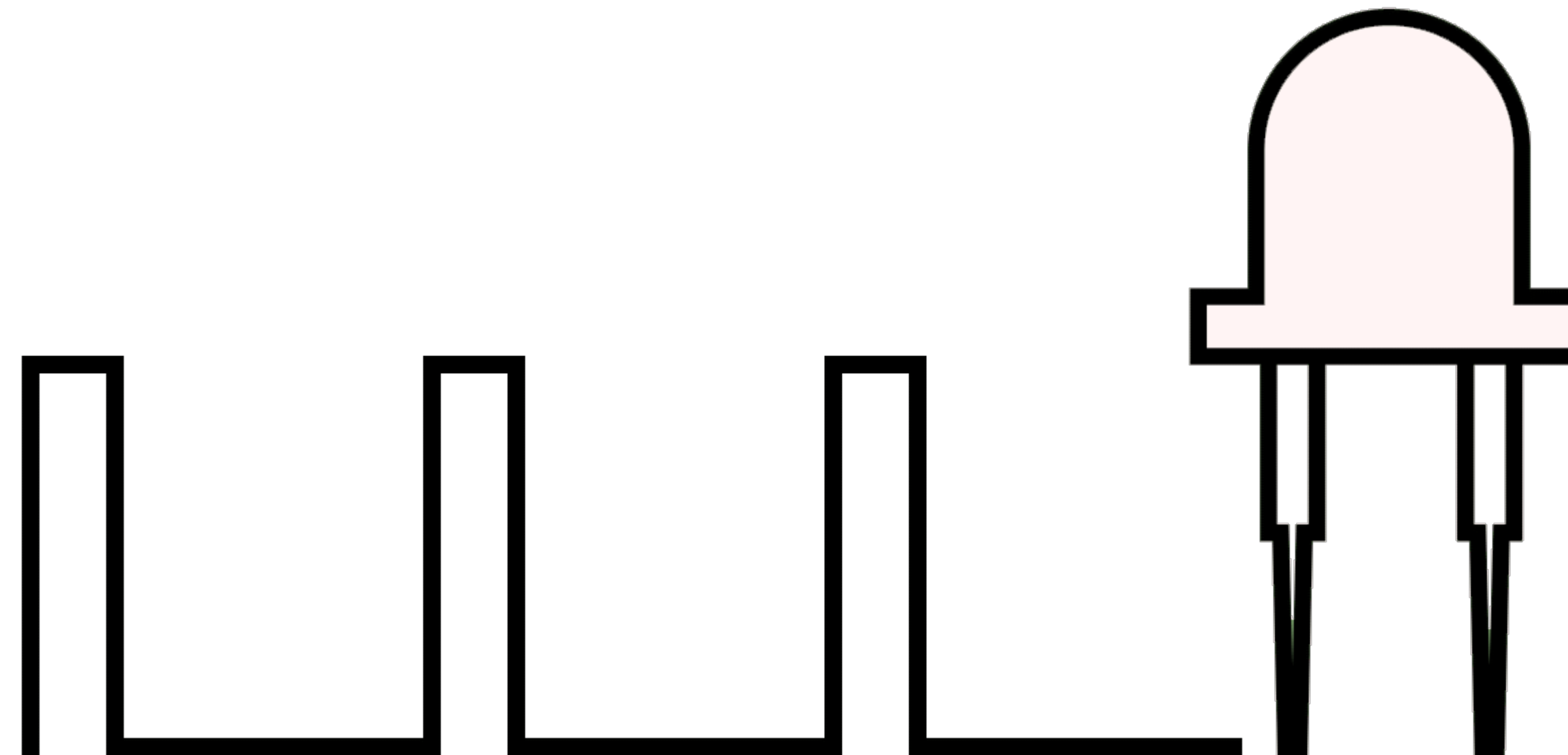


DC Motor: Speed Control

- In order to control the motor, a third type of signals has been introduced as a **hybrid** between the analog and digital signals.
- This signal type is known as the Pulse Width Modulation (PWM)

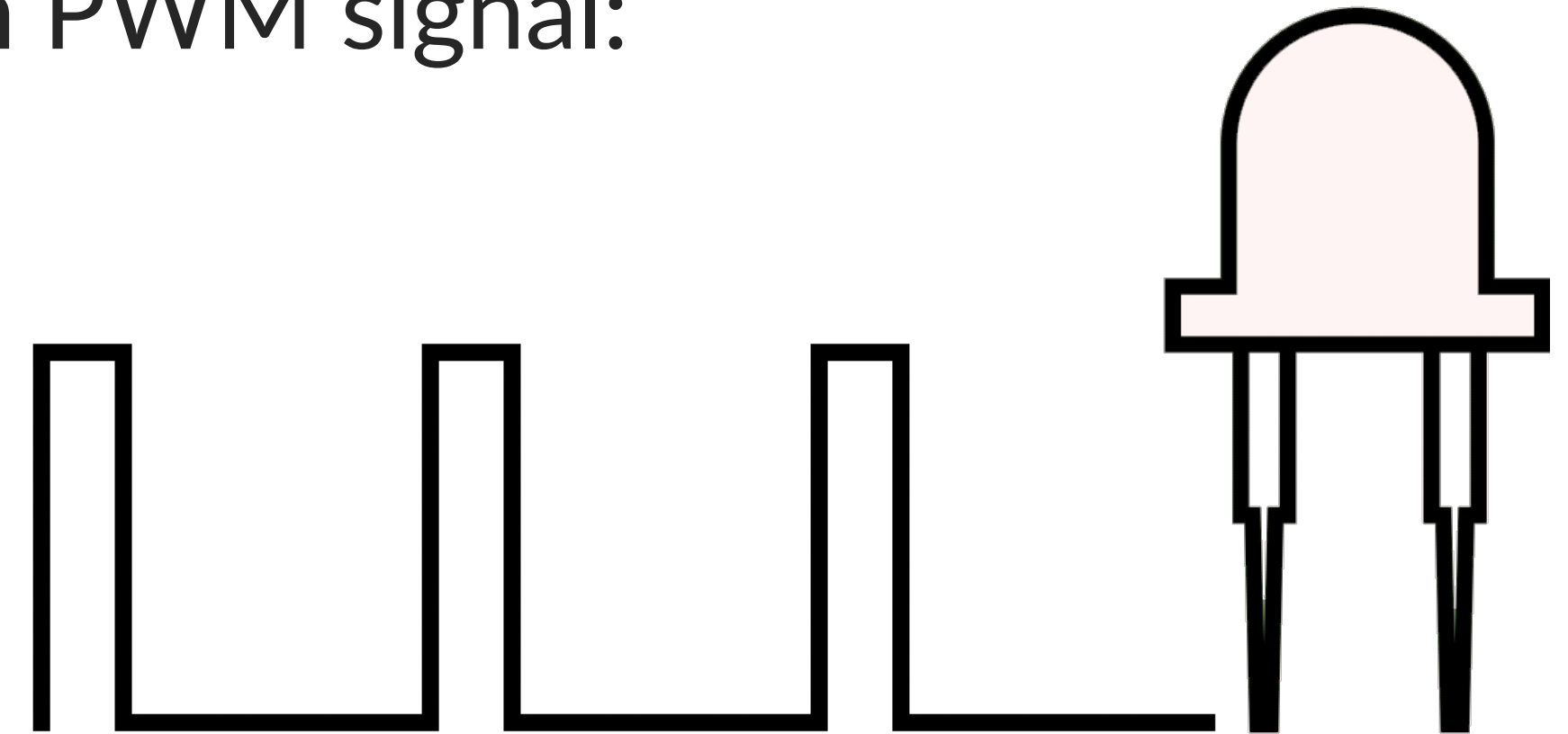
Pulse Width Modulation (PWM)

- The PWM is a digital signal that has a response of a digital one.
- That means we are generating a sequence of ones and zeros pulses yet we can get the effect of ranging the voltage from V_{min} to V_{max} .
- These pulses are generated for different time intervals and that what gives us the analog effect.



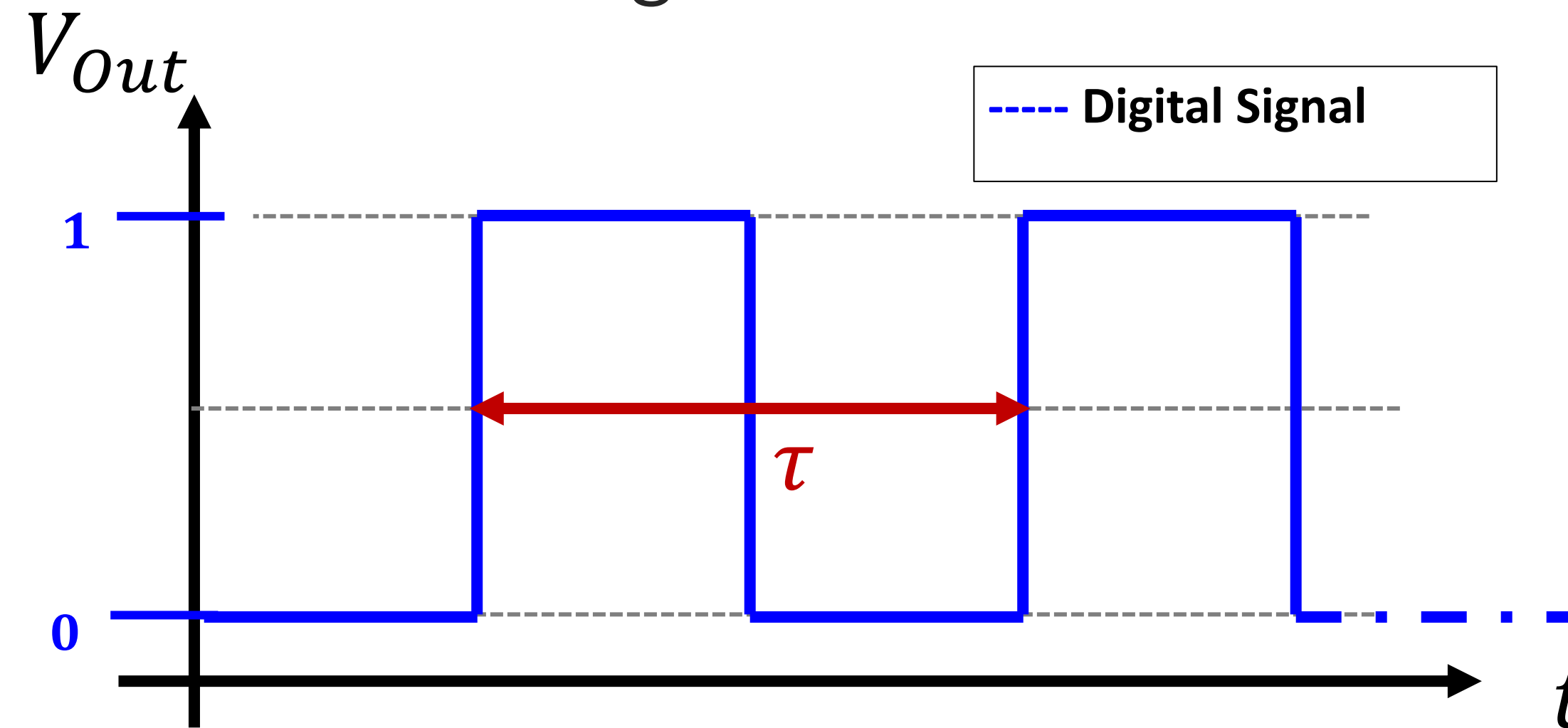
Pulse Width Modulation (PWM): Parameters

- There are three important parameters associated with PWM signal:
 - Duty Cycle of PWM
 - Frequency of PWM
 - Output Voltage of PWM signal



Pulse Width Modulation (PWM): Frequency

- The frequency of PWM determines how fast a PWM completes a period.
- The frequency of a pulse is shown in the figure above.



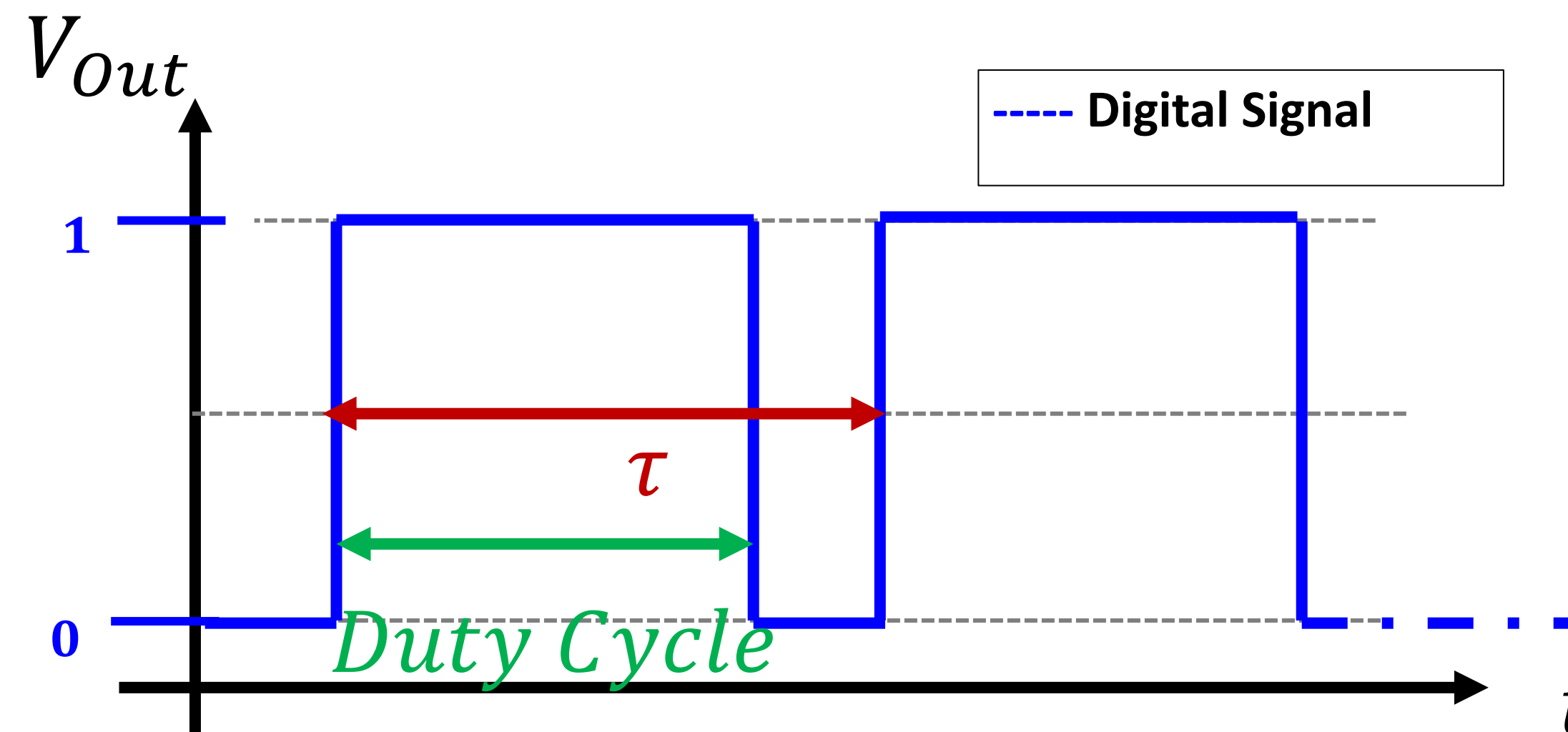
- The frequency of PWM can be calculated as follows:

$$Time\ Period\ (\tau) = Time_ON + Time_OFF$$

$$Frequency = \frac{1}{Time\ Period\ (\tau)}$$

Pulse Width Modulation (PWM): Duty Cycle

- The percentage of time for which the signal remains “ON” is known as the **duty cycle**.



- If the signal is always “ON,” then the signal must have a 100 % duty cycle.
- The formula to calculate the duty cycle is given as follows:

$$Duty\ Cycle(\%) = \frac{Time_{ON}}{Time_{ON} + Time_{OFF}}$$

- The average value of the voltage depends on the duty cycle.

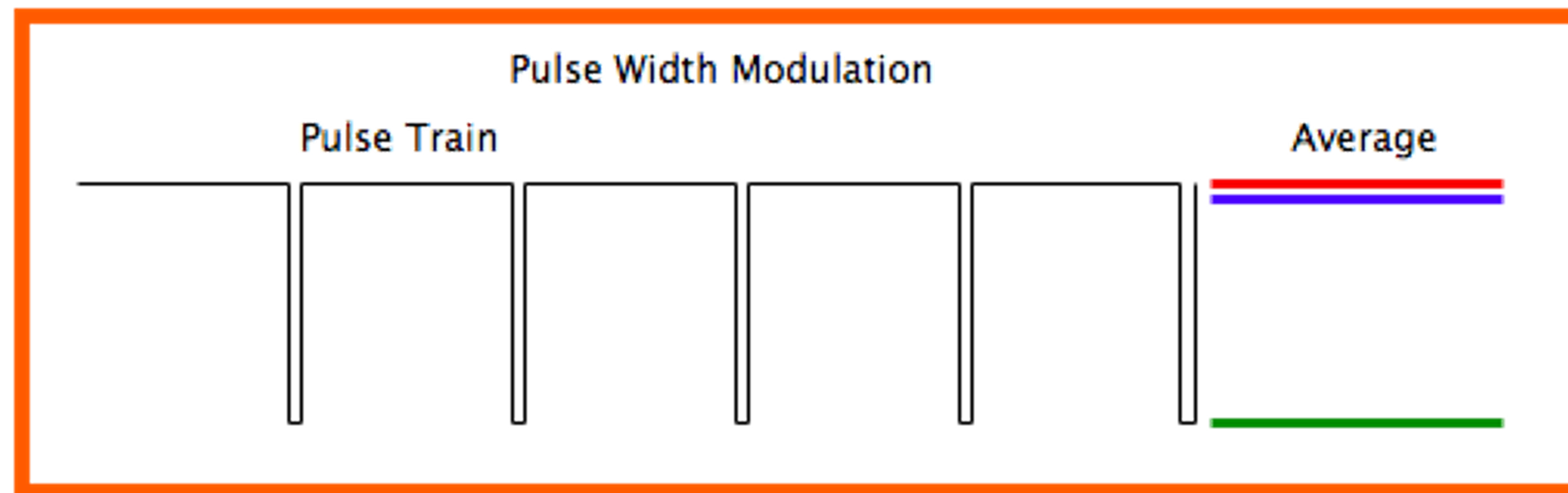
Pulse Width Modulation (PWM): Output Voltage

- The output voltage of the PWM signal will be **the percentage of the duty cycle**.

$$V_{out} = V_{max} \times Duty\ Cycle(\%)$$

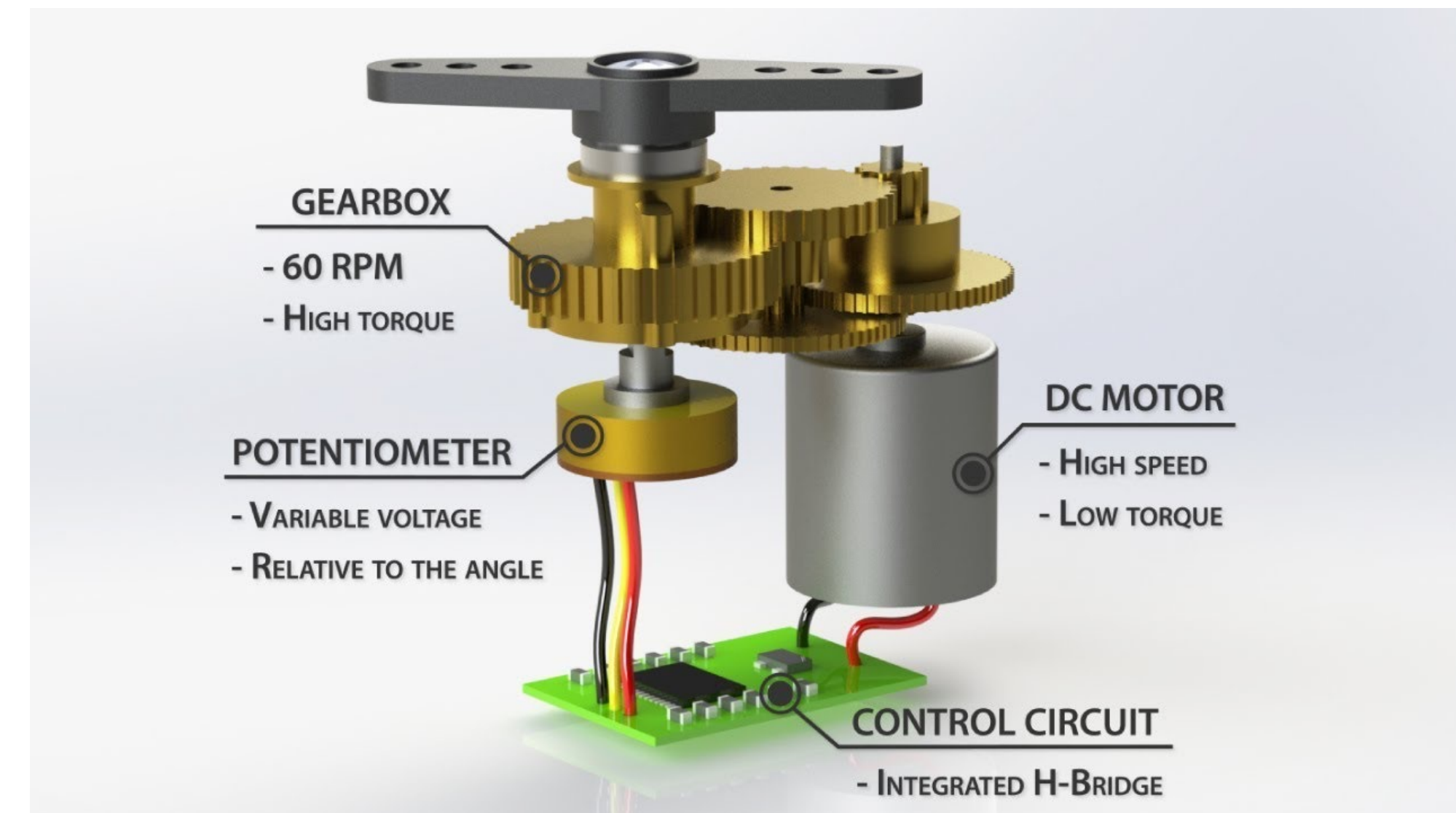
- For example

- For a 100% duty cycle, if the operating voltage is 5 V then the output voltage will also be 5 V.
- If the duty cycle is 50%, then the output voltage will be 2.5 V.
- If the duty cycle is 25%, then the output voltage will be 1.25 V.



Servo Motor

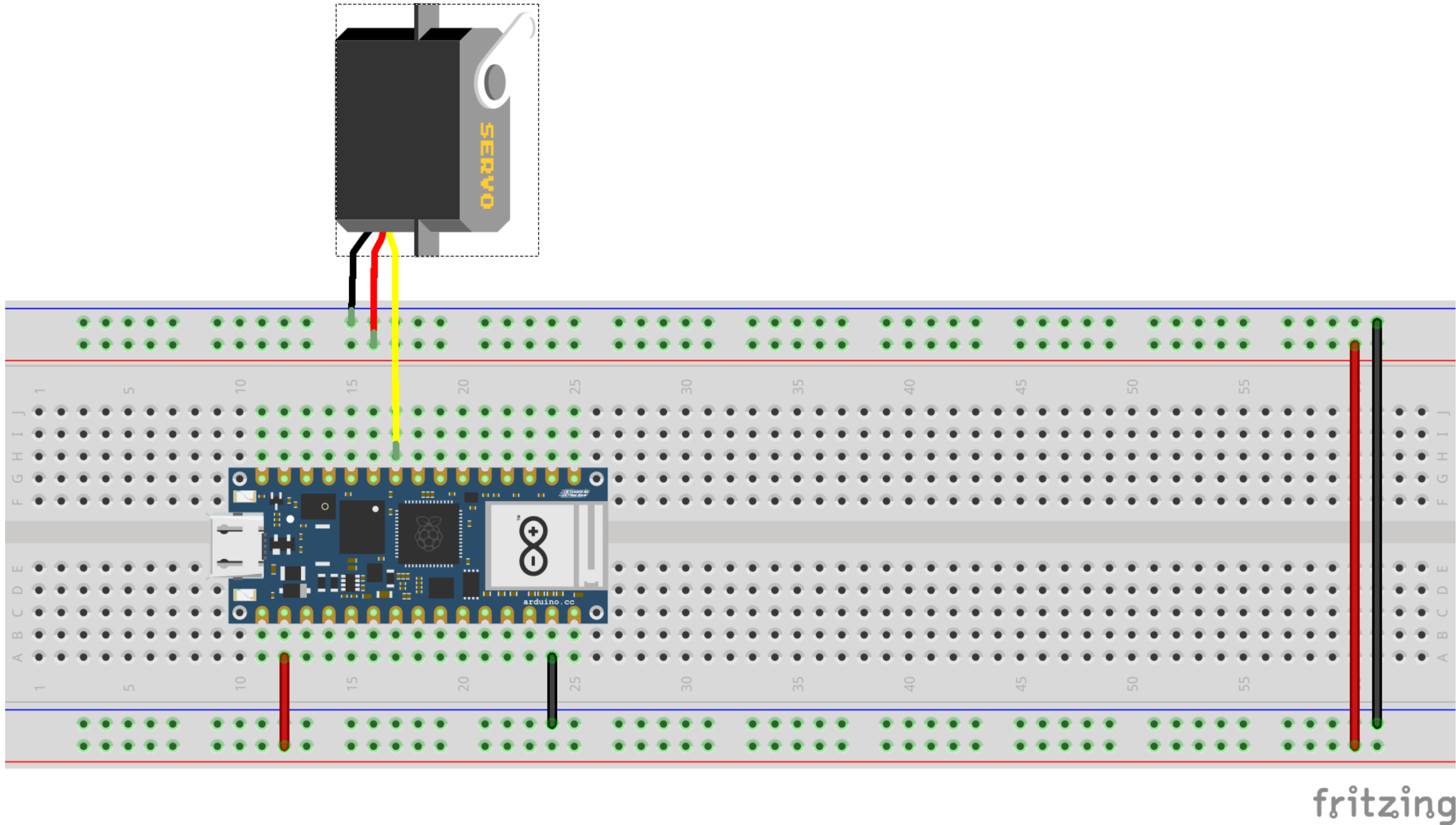
- A servo motor is a type of electric motor that is designed for precise control of position, speed, and acceleration.
- It is composed of a regular DC motor + a potentiometer for feedback on the current angle + a control circuit with integrated H-bridge that can ensure that the motor is at a precise angle position.
- Servo motors are commonly used in various applications, especially those that require high precision and accuracy, such as robotics, automation, CNC machinery, and remote-controlled systems.



Servo Motor

- Some of the key features and characteristics of servo motors are:
 - **Closed-Loop Control:** Servo motors are typically used in closed-loop control systems, which means they are equipped with feedback devices like encoders or resolvers. These feedback devices continuously monitor the motor's actual position and provide feedback to the control system. This allows the control system to make real-time adjustments to ensure the motor reaches and maintains the desired position or speed.
 - **Precise Position Control:** Servo motors are known for their exceptional precision when it comes to positioning. They can move to a specific angle or position with great accuracy, often down to fractions of a degree.

Servo Motor: Connection with μ C





ES Modeling & Design (2 Modules)

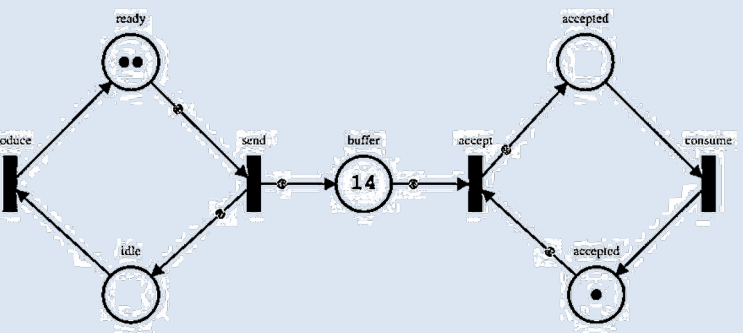
System Modeling

Let's go

Design Considerations

Power management and optimization

Reliability and fault tolerance



The Real-time Embedded System

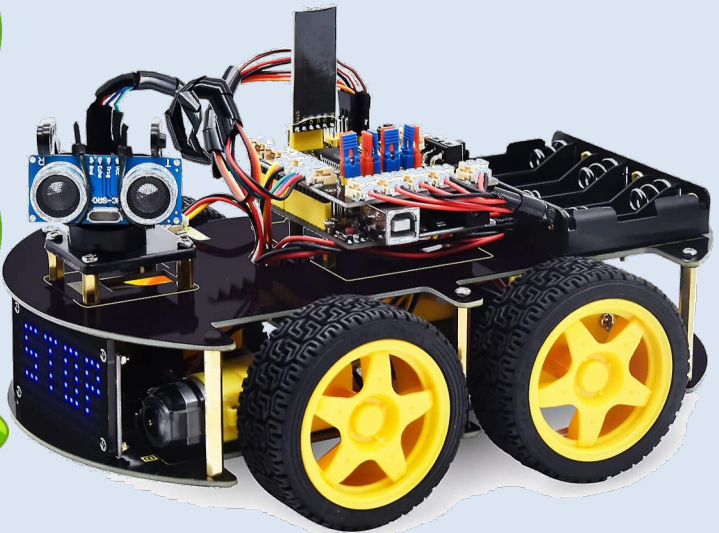
ES Hardware Components (4 Modules)

Microcontroller
Fundamentals

Embedded programming
languages

Embedded Hardware

Communication and
Networking



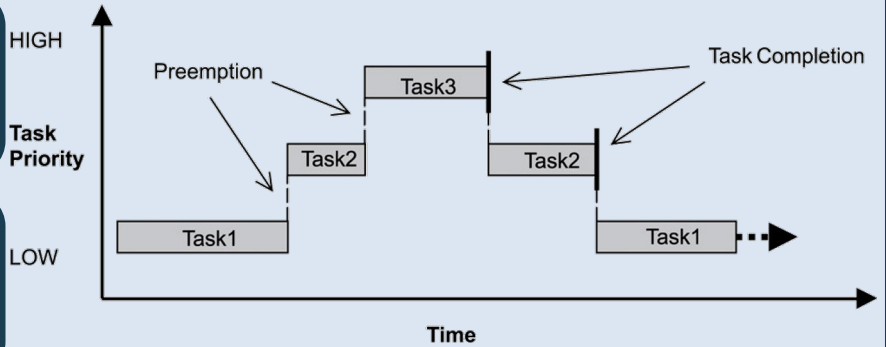
ES Software Components (1 Module)

Real-Time Systems

Multi-tasking

Scheduling

Resource Management



Embedded System Tools & Software Development (2 Modules)

Debugging techniques

Interrupts and exception
handling

Memory management



For Further Inquiries, Please
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Thank you for your attention!

See you next time 😊